ASGSR

Virall

A Technology-Agnostic Pipeline for Viral Biosurveillance on the International Space Station

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WHY THIS MATTERS

Why monitor viruses in space stations?

Months in a closed loop

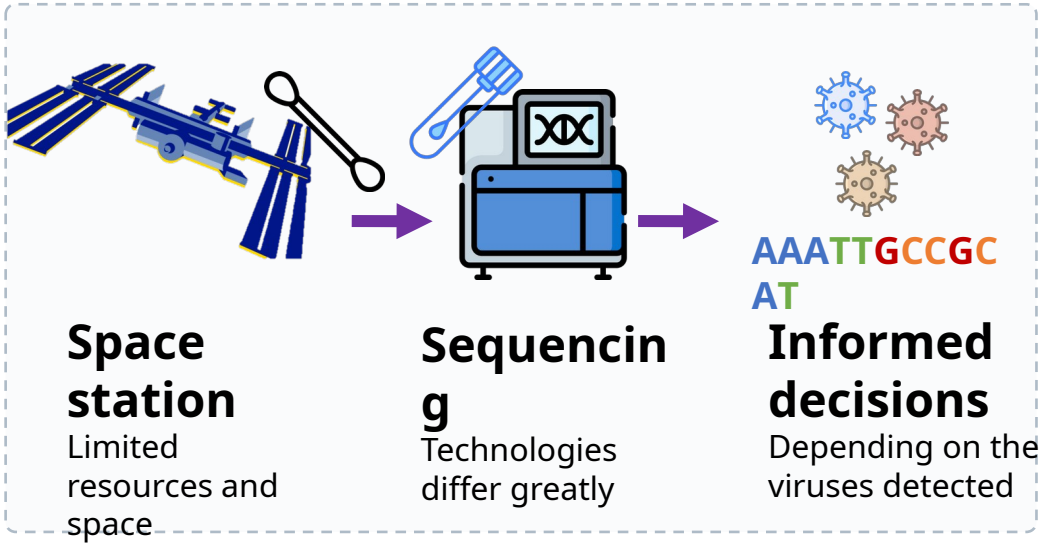
Crews share air, water, and surfaces for long periods. A closed habitat is a shared microbial environment.

Surfaces are reservoirs

What settles on a handrail, a water line, or a filter can persist long after the source is gone.

Early warning matters

Detecting a change early protects crew health and mission success — before anyone gets sick.



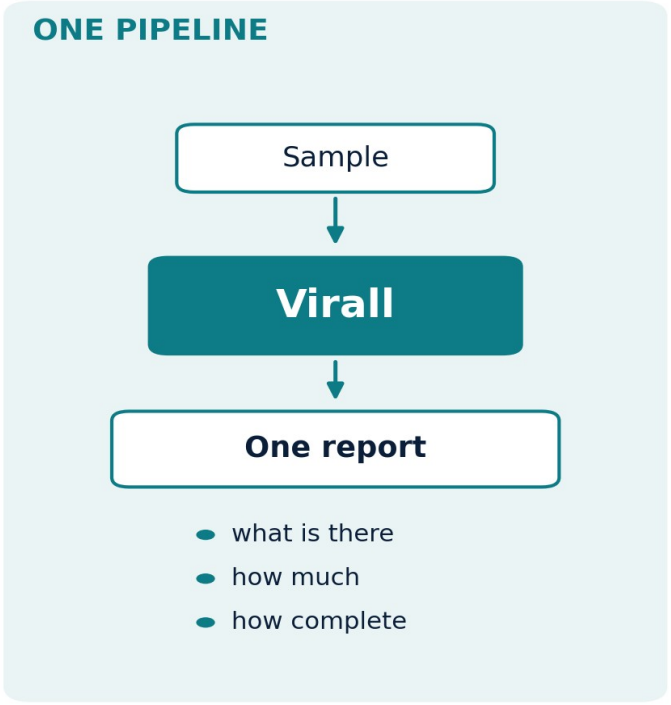
THE GAP

The tools are fragmented

Different treatment per sequencing technology
Short-read, long-read, and hybrid data often need different pipelines.

Detection is not enough
Finding a virus, counting it, and checking its genome quality live in separate tools.

Hard to compare missions
Different methods make it difficult to know what changed between flights.



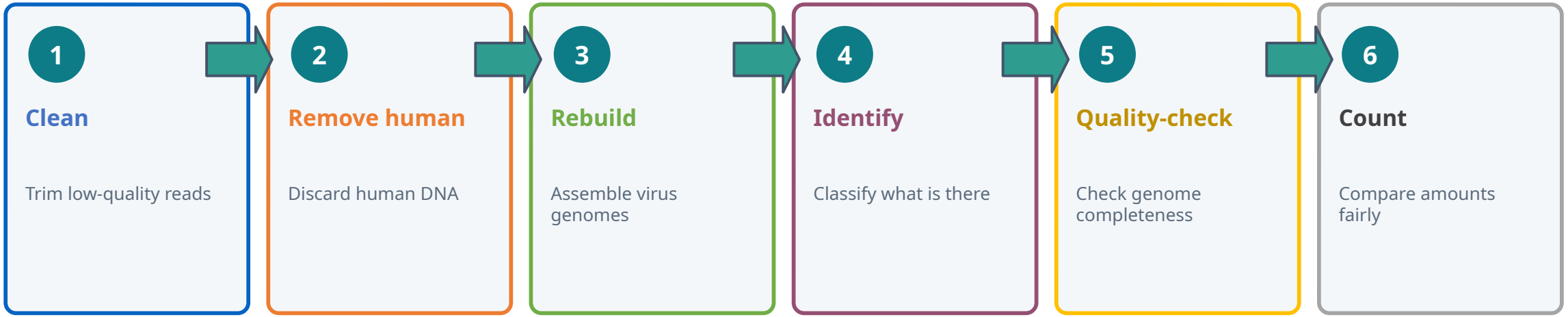
One reproducible tool to answer: what viruses are there, how much, how complete are they?

THE TOOL

What Virall does

Giving it a samplesheet and run:

```
nextflow run nextflow/main.nf -params-file nextflow/run_params.yaml -profile docker
```



Short & long reads · DNA & RNA · bulk & single-cell

One reproducible path from sample to comparable answer.

VALIDATION

Validate efficiency

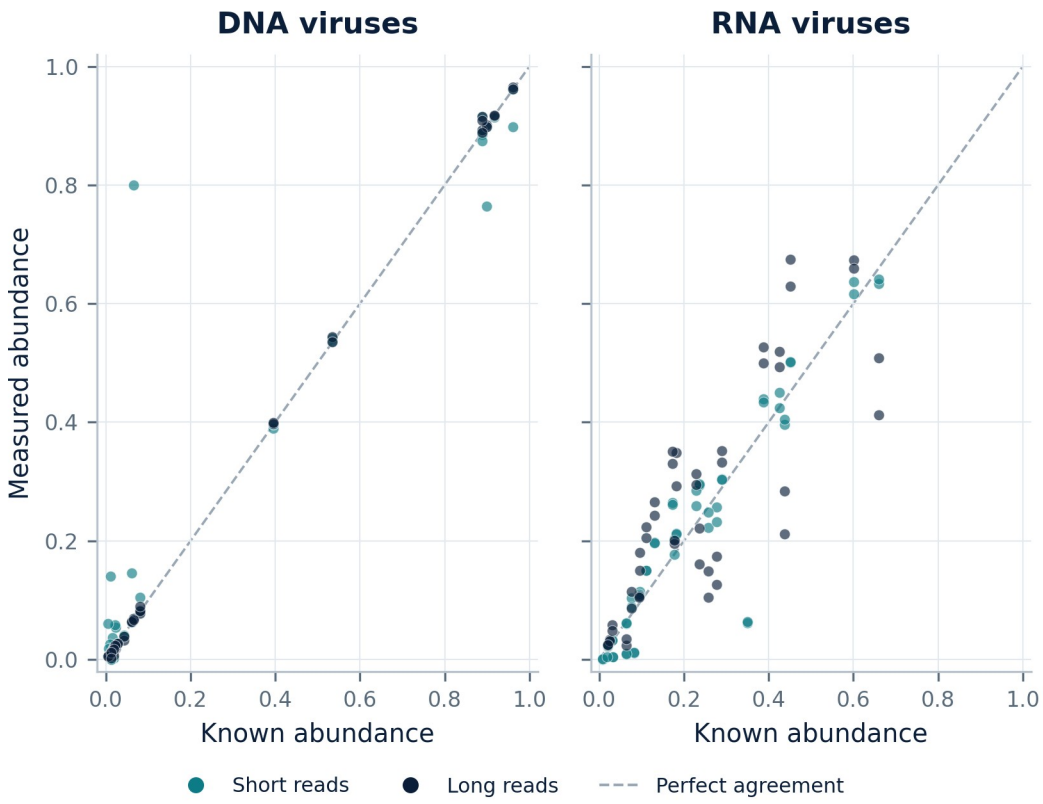
48 simulated communities

5 RNA + 4 DNA viruses
DNA (4): HSV-1 · cytomegalovirus · hepatitis B · adenovirus C
RNA (5): SARS-CoV-2 · influenza A · measles · hepatitis C · rotavirus

4 platforms: Illumina · IonTorrent · Nanopore · PacBio

Fit between known and measured viral abundance

Higher for DNA viruses



Spaceflight demonstration: Virome of ISS

Detect viral genomes in the swabs over multiple missions and differentiate infections viruses vs non-infectious

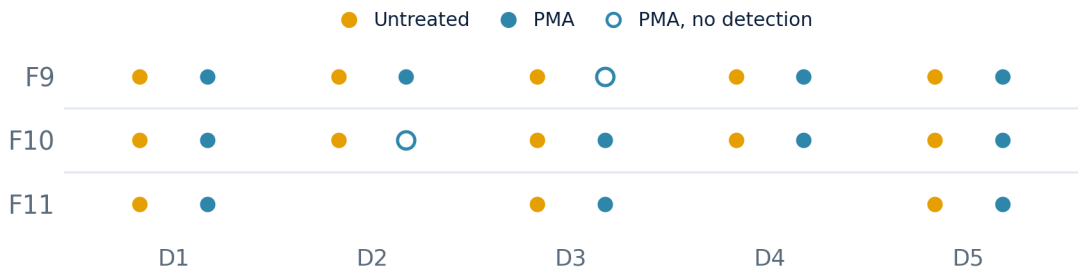
Propidium monoazide (PMA): a filter for intact virus

PMA binds DNA in broken virus particles and blocks it from being sequenced.

Intact particles are unaffected, so we read what is intact, not just what is present.

26 paired swabs over **3** flights

13 treated + 13 untreated



13 paired swabs across three flights; 2 PMA samples had no detectable viruses

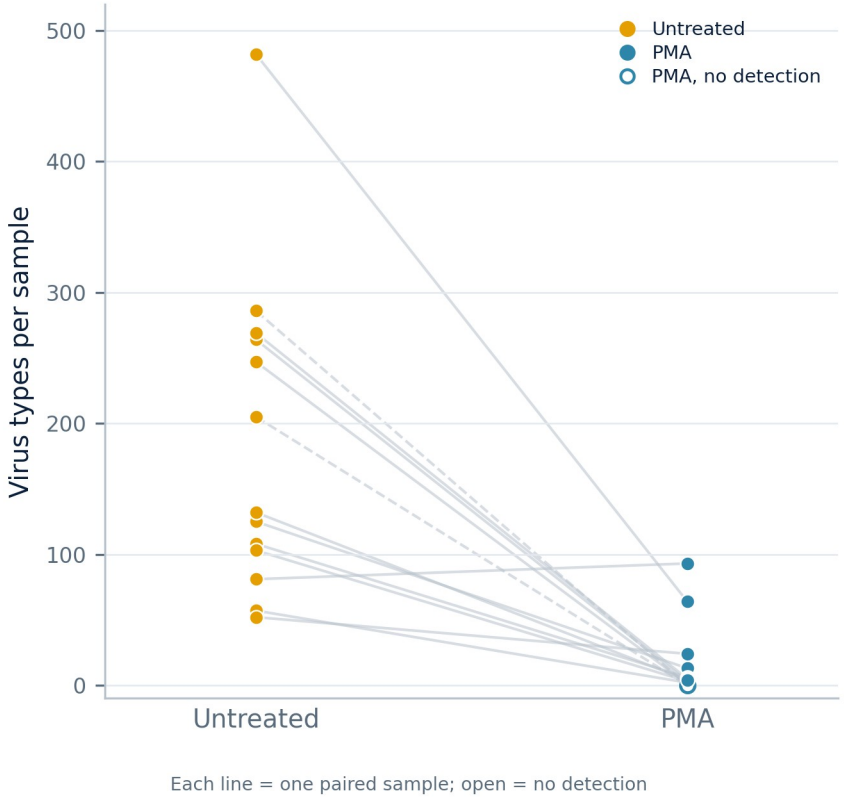
Most viral DNA is not viable

Viral species count

Before PMA **After PMA**

52-482 → 0-93

Most of the viral DNA on these surfaces is free or relic material and **not infectious virus**.



Each line = one paired sample; open = no detection

A persistent bacterial-virus core remains

128 total species survived PMA

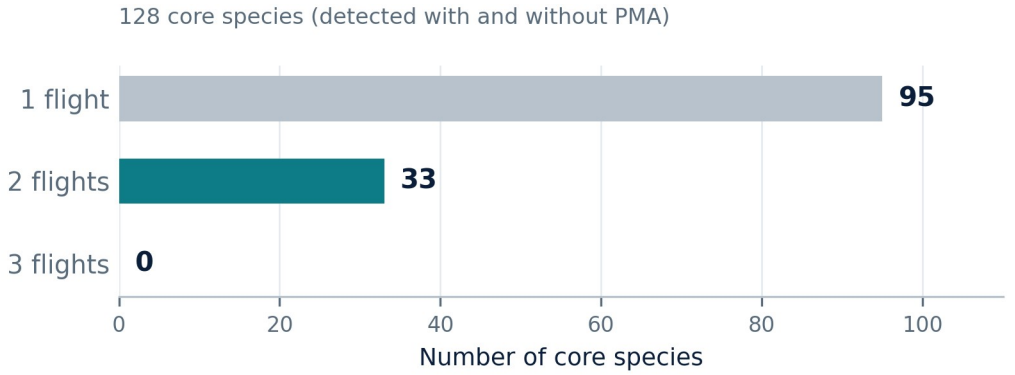
Membrane not damaged

33 seen in PMA in 2 flights

0 seen in all 3 flights

Virome of ISS surfaces

A stable functional core with dynamic species turnover

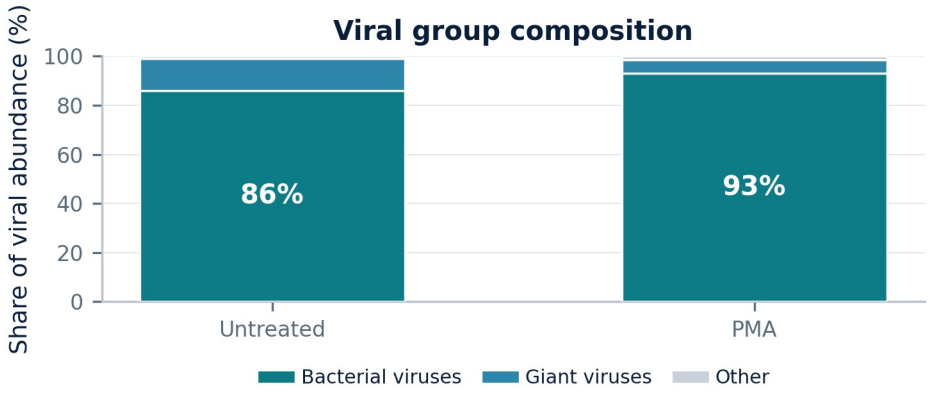


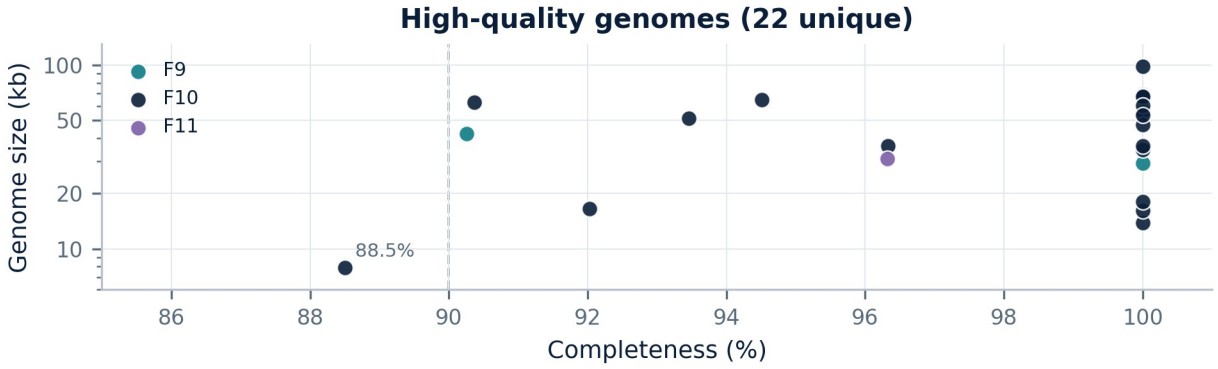
What survives and what we recovered

bacterial-virus share
86% → 93%

giant-virus share
13% → 6%

22 unique high- quality genomes

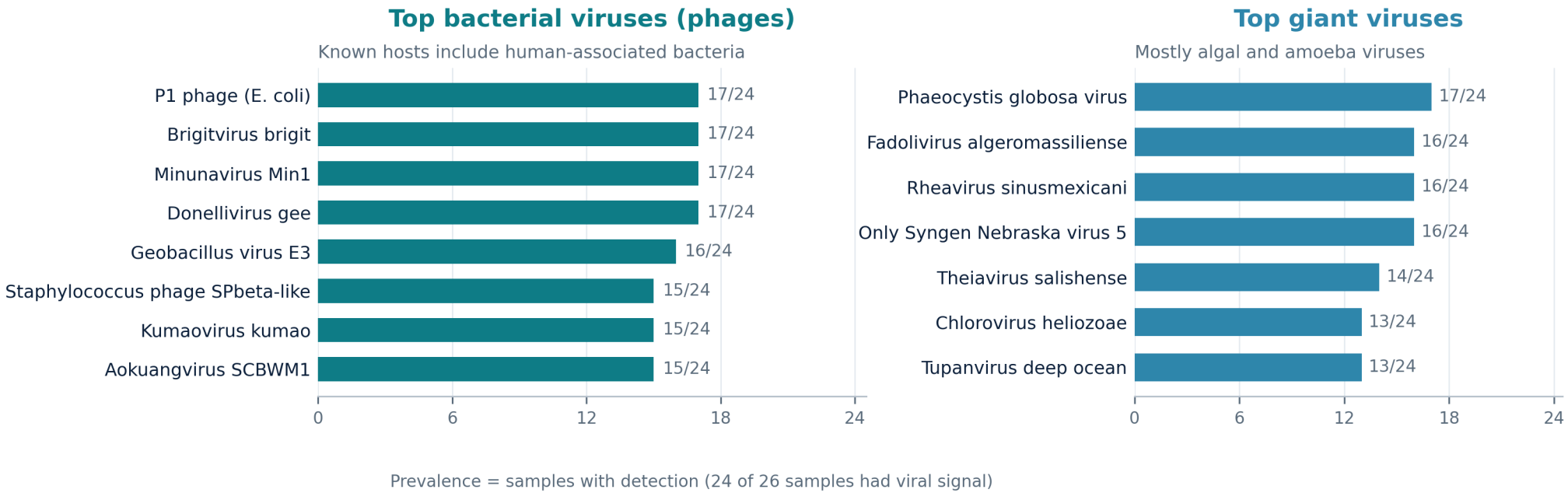




All high-quality records came from untreated samples; 16 of 26 from one deep F10 sample.

Viruses of ISS surfaces

Most prevelent species



Prevalence = samples with detection (24 of 26 samples had viral signal)

Several prevalent phages infect human-associated bacteria

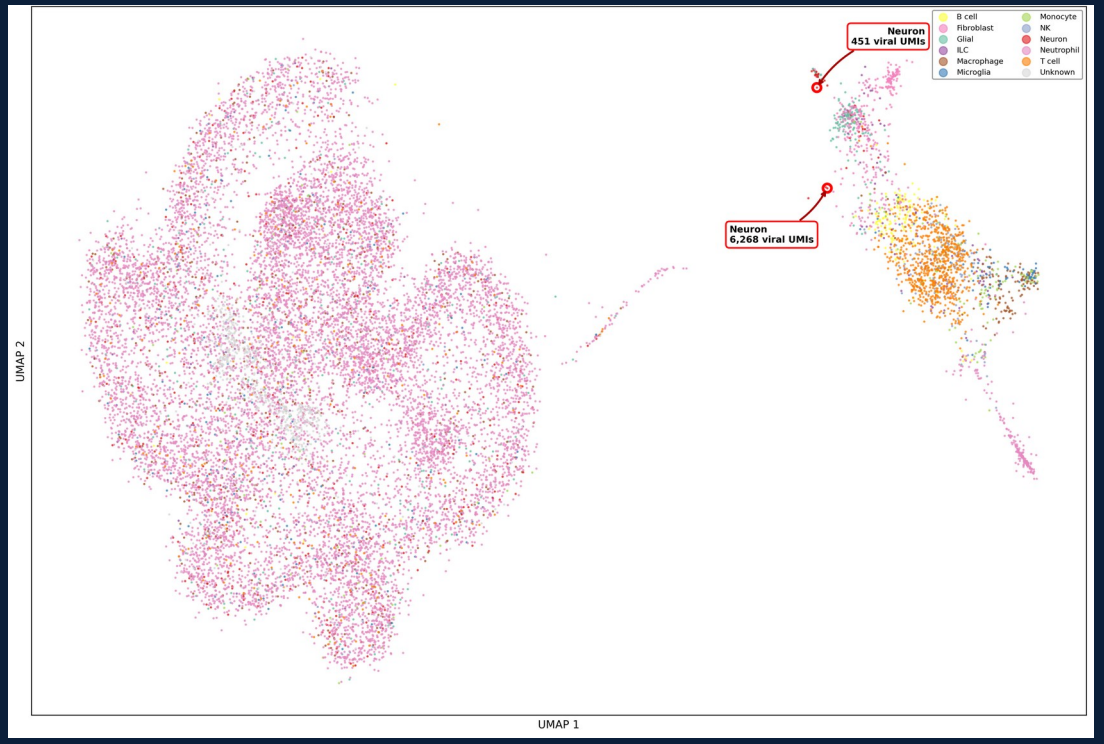
DIFFERENT DATASET · CAPABILITY DEMO

Beyond bulk: single-cell capability

SEPARATE STUDY

2 **infected neurons (trigeminal ganglions)**
among 17,014 cells

HSV-1 mouse sensory-neuron single-cell data (GSE225839). Virall can pick out rare infected cells from thousands of uninfected ones.



WHAT IT MEANS

What this means for spaceflight

Monitor environmental viruses easily

PMA should be standard in environmental surveillance.

One tool, many missions

Comparable results across sequencers and flights.

Define the “normal” virus community

The persistent bacterial-virus core is a baseline for station health.

TAKE-HOME

Three things to remember

- Virall is validated, works with any sequencer, and is built for spaceflight.
- Spacecraft virus communities are mostly non-viable DNA , with a persistent core of bacterial viruses.
- Surveillance should measure intact virus, not just viral DNA.

Explore & docs

virall-explorer.pages.dev

SCAN TO EXPLORE

Thank you, questions?

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